EgoLife: A Week-Long Egocentric Benchmark and EgoButler for Personalized Life Assistance

Anonymous Submission

Abstract

Building an egocentric life assistant requires both a benchmark that measures personalized reasoning over real daily life and a method that can retrieve and reason over extremely long multimodal context. We present EgoLife, a week-long egocentric dataset collected from six cohabiting participants in an instrumented house, totaling 266 hours of synchronized first-person video with audio and additional ego-exo signals, and EgoLifeQA, a 3,000-question multiple-choice benchmark designed for life-assistant queries. EgoLifeQA spans five task families, from entity and event recall to habit inference, social relation understanding, and task management, and explicitly annotates certificate length to measure how much historical context each question requires. The benchmark is challenging: 2,003 of 3,000 questions require more than two hours of context, and strong proprietary and open-source baselines remain near the mid-30s in average accuracy. To address this setting, we introduce EgoButler, a retrieval-augmented system that combines EgoGPT, an egocentric omni-modal captioning model built on LLaVA-OneVision with an audio branch, and EgoRAG, a hierarchical memory and retrieval pipeline over clip-, hour-, and day-level summaries. Experiments show that Day-1 personalization improves EgoGPT on EgoLifeQA from 33.1 to 36.0 average accuracy and especially helps identity- and relation-centric questions, while EgoRAG substantially improves long-horizon reasoning, raising accuracy from 26.8 to 38.9 on 6–24 hour questions and from 25.0 to 35.4 on questions requiring more than 24 hours of look-back. The results also expose remaining bottlenecks: short-context questions do not benefit from retrieval, and high-quality visual-audio captions remain critical for downstream performance.

1. Introduction

Egocentric perception is a natural substrate for always-on assistants because it captures what a wearer sees, hears, and does over time. Yet the current ecosystem of egocentric datasets and benchmarks remains mismatched to that goal. Large-scale datasets such as EPIC-KITCHENS, Ego4D, and Ego-Exo4D have enabled substantial progress in first-person understanding, but they focus on short clips, task-specific scenarios, or skilled activities rather than continuous daily life in a shared social environment [1–3]. Surveys of the field likewise identify long-term memory, personalization, and socially grounded assistance as open problems for future egocentric systems [4, 5].

The evaluation gap is equally important. Recent egocentric QA benchmarks, including EgoSchema, EgoPlan-Bench, EgoThink, and MM-Ego, are valuable but mostly operate on single clips or sub-hour histories [6–9]. Even recent long-video benchmarks emphasize hour-scale reasoning or web videos rather than week-long, personalized, multi-person daily life [10, 11]. As a result, they under-measure the kinds of questions a practical life assistant must answer: who last used a tool, what habit usually accompanies an activity, when a recurring event happened, or which friend most often joins a social routine.

This paper addresses both sides of the problem. First, we introduce **EgoLife**, a week-long multimodal egocentric dataset collected in an instrumented shared house, and **EgoLifeQA**, a benchmark of 3,000 multiple-choice questions with explicit certificate-length annotations. EgoLifeQA contains five task types: *EntityLog*, *EventRecall*, *HabitInsight*, *RelationMap*, and *TaskMaster*. The benchmark is intentionally built around personalized, socially grounded, and long-horizon questions. Second, we introduce **EgoButler**, a system for long-horizon egocentric QA that combines omni-modal clip understanding with hierarchical retrieval over week-long memory. EgoButler uses **EgoGPT** to convert egocentric video and audio into first-person captions, and **EgoRAG** to build multi-level summaries and retrieve evidence for answer generation.

Our experiments highlight both progress and remaining difficulty. On established egocentric benchmarks, EgoGPT is competitive with strong open and proprietary models and achieves the best result in our comparison on EgoSchema. On EgoLifeQA, adding Day-1 personalization improves average accuracy from 33.1 to 36.0, with especially large gains on EntityLog and RelationMap. Hierarchical retrieval is most beneficial for long-horizon questions: accuracy rises by more than 10 points for questions with certificate length above six hours, while slightly decreasing on the shortest

Figure 1. **EgoLife overview.** EgoLife records six people living in the same house over seven days using egocentric multimodal sensors and synchronized exocentric cameras. The benchmark targets assistant-style questions that require identity tracking, event recall, habit discovery, relation understanding, and task management over ultra-long context.

bucket. These mixed results are important. They show that retrieval and personalization matter, but also that week-long egocentric assistance remains far from solved.

Our contributions are threefold. (1) We introduce EgoLife, a 266-hour week-long ego-exo dataset with dense transcripts and visual-audio captions for continuous daily life. (2) We introduce EgoLifeQA, a long-context benchmark with certificate-length metadata and five assistant-oriented task families. (3) We present EgoButler, a retrieval-augmented pipeline that couples omni-modal egocentric captioning with hierarchical memory retrieval, and we provide a detailed analysis of where it helps and where it still fails.

2. Related Work

Egocentric datasets. Egocentric vision has been driven by datasets spanning daily activities, cooking, procedural tasks, and skilled activities [1–3, 12, 13]. Earlier datasets such as ADL, Charades-Ego, EGTEA Gaze+, and HoloAssist helped establish first-person recognition, narration, and assistant-like interaction settings [14–17]. However, most existing collections are clip-based, activity-centric, or lack sustained interpersonal dynamics. EgoLife differs by emphasizing continuous multi-day daily life, synchronized ego-exo capture, and questions that require recalling events and relationships across a shared week.

Egocentric QA and long-video evaluation. Benchmarking long-horizon reasoning has recently expanded through EgoSchema, EgoPlan-Bench, EgoThink, MM-Ego, HourVideo, LongVideoBench, and Video-MME [6–11, 18]. These benchmarks expose the difficulty of long-context video understanding, but they primarily evaluate clip-level, hour-scale, or non-personalized reasoning. EgoLifeQA is complementary: it centers on life-assistant questions over a single user's week-long memory and introduces certificate length as a benchmark primitive for multi-day personalization.

Egocentric multimodal models and retrieval. Recent video-language models and omni-modal systems, including LLaVA-OneVision, LLaVA-NeXT-Video, LongVA, Oryx, Qwen2-Audio, GPT-4o, and Gemini-1.5-Pro, have improved video understanding and multimodal interaction [19–25]. Egocentric-specific pretraining and instruction tuning have also advanced through EgoVLPv2 and MM-Ego [9, 26]. For long-form reasoning, hierarchical summarization and memory mechanisms are increasingly common in video captioning and long-video understanding [27–30]. EgoButler builds on this direction, but targets the more specific regime of week-long egocentric QA with personalized and socially grounded queries.

Table 1. **Related Work for EgoLife Dataset - Overview of Egocentric Datasets.** Modality abbreviations are **V**: video, **G**: gaze, **I**: IMU, and **3D**: 3D scans. EgoLife stands out for its ultra-long egocentric footage and rich interpersonal interactions.

Benchmark	Domain	Modality	#Captions	Size (hrs)	#Clips	Dur./Clip	Multiview	Interpersonal Dynamics
EPIC-KITCHENS [1]	Kitchen	V	20K+	100	700	8.5 min	✗	✗
Ego4D [2]	Daily Activities	V+G+I+3D	3.85M	3,670	9,645	22.8 min	✗	✗
EgoExo4D [3]	Skilled Activities	V+G+I+3D	500K+	1,286	5,035	1 to 42 min	✓	✗
EgoExoLearn [12]	Task Execution	V+G+I	-	120	432	13.4 min	✓	✗
EgoPet [13]	Animal Actions	V+G+I	-	84	6,646	45.5 sec	✗	✗
EgoLife	Daily Life	V+G+I+3D	400K+	266	6	44.3 h	✓	✓

Table 2. **Related Work for EgoLifeQA Benchmark.** EgoLifeQA is distinguished by its ultra-long video footage and certificate length, enabling tasks such as habit discovery and relational interaction pattern analysis. **Note on Dur./Clip:** A clip is defined as a session with narrative continuity. For EgoLife, this value is derived from 266 hours of retained footage distributed across six participants.

Dataset	Source	#QAs	Size (hrs)	#Clips	Dur./Clip	Certificate Length [6]	
						Below 2h	Over 2h
EgoSchema [6]	Ego4D	5,063	250	5,063	3 min	5,063	0
EgoPlan-Bench [7]	Ego4D & EpicKitchen	4,939	-	4,939	-	4,939	0
EgoThink [8]	Ego4D	700	-	595	-	700	0
EgoMemoria [9]	Ego4D	7,026	-	629	30 s to 1 h	7,026	0
HourVideo [10]	Ego4D	12,976	381	500	20 min to 2 h	12,976	0
EgoLifeQA	**EgoLife**	3,000	266	6	44.3 h	997	2,003

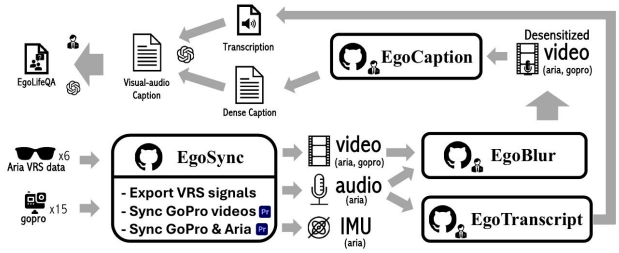


Figure 2. **Data processing pipeline.** EgoLife synchronizes Aria and GoPro streams, applies desensitization, produces transcripts and dense captions, and converts them into visual-audio captions that support EgoLifeQA construction.

3. EgoLife and EgoLifeQA

3.1. Dataset Overview

EgoLife is collected in an instrumented shared house using six Project Aria devices for egocentric capture, 15 GoPro cameras for exocentric coverage, and two mmWave devices in the common area [31]. The retained dataset contains 266 hours of egocentric recordings across six participants over seven days, corresponding to 44.3 hours per participant on average. Compared with prior egocentric datasets, EgoLife emphasizes ultra-long duration, multi-view synchronization, and rich interpersonal interaction rather than isolated task execution, as summarized in Table 1.

The activity distribution is broad. The seven-day timeline includes 14 high-level categories, led by social interactions (183 segments), housekeeping (145), cooking (86), shopping (74), dining (67), and party-related activities (64), with additional arts and craftwork, leisure, games, music and dance, outings, setup, meetings, and commuting. This diversity matters because a life assistant must reason over both routine and socially contingent events, rather than over a narrow procedural domain.

3.2. Data Processing and Annotation

Figure 2 summarizes the data pipeline. EgoSync aligns Aria and exocentric streams; desensitization is handled before downstream processing; and transcripts are produced from egocentric audio using Whisper-style ASR with diarization support [32, 33]. The pipeline then creates dense captions and final visual-audio captions that describe each clip from the wearer's first-person perspective, using GPT-4o-assisted consolidation with human review [24]. This produces the text substrate used both for benchmark construction and for retrieval-based reasoning.

The dataset also supports model development through **EgoIT-99K**, an instruction-tuning corpus built from nine public egocentric datasets. As shown in Table 3, EgoIT-99K contains 99.48K training instances over 43.16 hours of video, including captioning, multi-choice QA, multi-round QA, and image QA supervision. Audio is available for 686 of 1,529 videos, enabling multimodal adaptation rather than vision-only instruction tuning.

3.3. Benchmark Design

EgoLifeQA contains 3,000 multiple-choice questions covering five task families that reflect life-assistant use cases. The benchmark includes 753 EntityLog questions, 751 EventRecall questions, 754 RelationMap questions, 378 HabitInsight questions, and 364 TaskMaster questions. These categories intentionally combine factual recall with social and behav-

Table 3. **Dataset Composition of EgoIT-99K.** We curated 9 classic egocentric video datasets and leveraged their annotations to generate captioning and QA instruction-tuning data for fine-tuning EgoGPT, building on the LLaVA-OneVision base model [19]. **#AV** means the number of videos with audio used for training. QAs include multiple types - **VC**: Video Captioning, **AVC**: Audio-Video Captioning, **MCQ**: Multiple Choice Questions, **MRC**: Multi-Round Questions, **IQA**: Image Question-Answering.

Dataset	Duration	#Videos (#AV)	#QA	QA Type
Ego4D [2]	3.34h	523 (458)	1.41K	VC, AVC, MCQ, MRC
Charades-Ego [15]	5.04h	591 (228)	18.46K	VC, AVC, MRC
HoloAssist [17]	9.17h	121	33.96K	VC, MCQ, MRC, IQA
EGTEA Gaze+ [16]	3.01h	16	11.20K	VC, MCQ, MRC, IQA
IndustReal [34]	2.96h	44	11.58K	VC, MCQ, MRC, IQA
EgoTaskQA [35]	8.72h	172	3.59K	VC, MCQ, MRC
EgoProceL [36]	3.11h	18	5.90K	VC, MCQ, MRC, IQA
Epic-Kitchens [1]	4.15h	36	10.15K	VC, MCQ, MRC, IQA
ADL [14]	3.66h	8	3.23K	VC, MCQ, MRC, IQA
Total	**43.16h**	**1529 (686)**	**99.48K**	

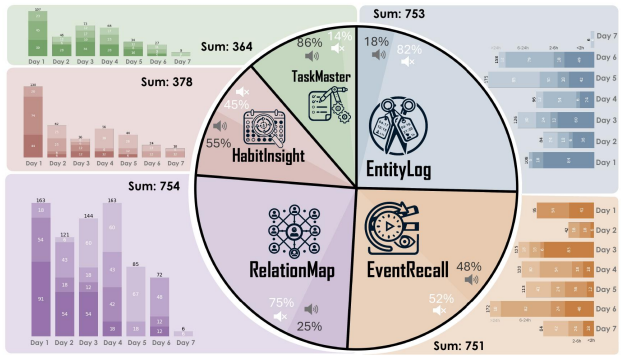


Figure 3. **EgoLifeQA statistics.** The benchmark covers five assistant-oriented task types, spans all seven days, and contains a large fraction of questions whose certificate length exceeds two hours.

ioral inference. EntityLog asks about people, objects, and attributes over time; EventRecall targets specific past happenings; HabitInsight probes recurring routines; RelationMap focuses on interpersonal patterns; and TaskMaster tracks ongoing plans, shopping, and household tasks.

Certificate length is a central design choice. Following EgoSchema's notion of temporal certificates [6], EgoLifeQA annotates how much past context is needed to answer each question. Table 2 and Figure 3 show that 2,003 of 3,000 questions require more than two hours of look-back, versus none in prior egocentric QA benchmarks listed in our comparison. The distribution is therefore not just longer in raw duration; it is longer in the amount of history genuinely needed for correct reasoning.

3.4. Evaluation Protocol

EgoLifeQA is evaluated with multiple-choice accuracy, reported overall and by question category. We also analyze performance by certificate-length bucket: below 2 hours, 2–6 hours, 6–24 hours, and above 24 hours. This split exposes whether a method truly benefits from long-horizon memory, rather than merely exploiting priors or short clips. The benchmark is designed so that evidence retrieval is part of the reasoning problem: successful systems must identify relevant parts of a week-long record before selecting an answer.

4. EgoButler

EgoButler is a two-stage system designed for personalized QA over week-long egocentric memory. It separates clip understanding from long-horizon retrieval. EgoGPT converts raw multimodal clips into textual memory units, while EgoRAG organizes and queries those units across multiple temporal scales.

4.1. EgoGPT: Omni-Modal Egocentric Captioning

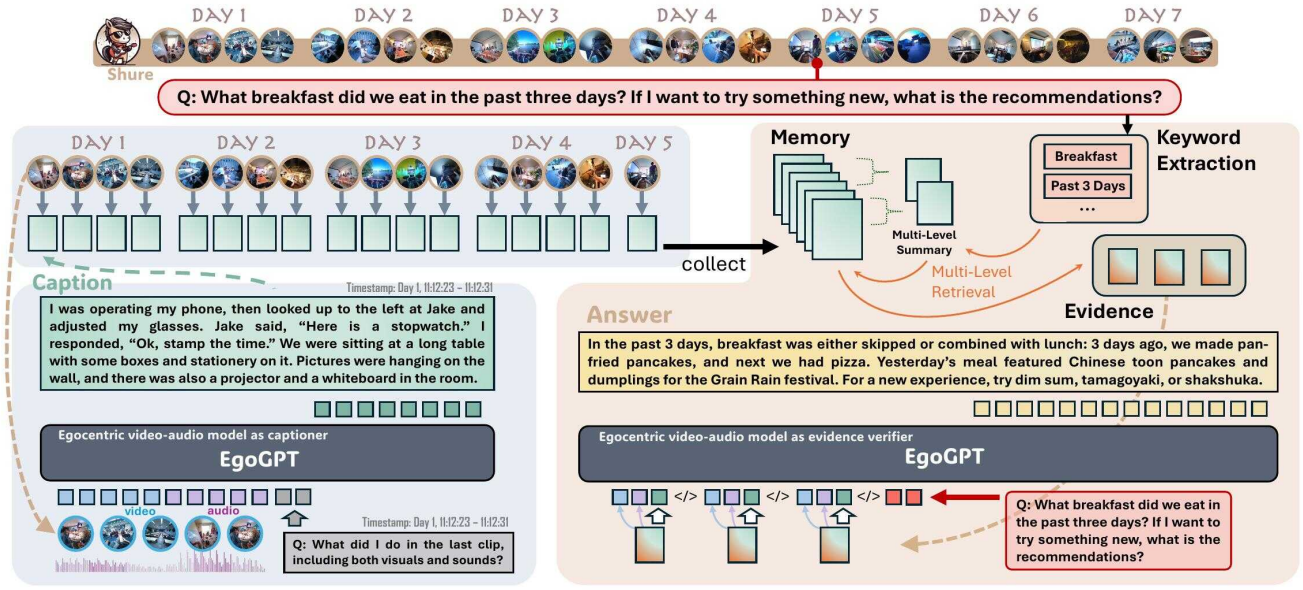
EgoGPT is built on the 7B LLaVA-OneVision backbone [19] and extends it with an audio branch. In the released implementation, video frames are sampled at 1 FPS, audio is converted to log-Mel features with Whisper preprocessing, and both modalities are injected through dedicated image and speech tokens. The model is prompted to caption clips from a first-person perspective, producing descriptions that preserve actions, objects, speech content, and interacting identities when possible.

Training uses EgoIT-99K, followed by an additional Day-1 personalization stage. The Day-1 setting adapts the model to the people, routines, and object identities observed during the first day of a participant's recording. This is a pragmatic form of personalization: it does not solve full life-long adaptation, but it allows us to test whether exposing the system to user-specific context improves later retrieval and QA.

4.2. EgoRAG: Hierarchical Memory and Retrieval

The retrieval component is derived directly from the provided implementation. First, EgoGPT captions each temporal segment and stores the resulting sentences in a Chroma database using OpenCLIP text embeddings. Each sentence retains metadata including day, start time, end time, and source video path. Second, the system builds a hierarchical memory by summarizing low-level captions into 10-minute events, then hourly summaries, and finally day-level summaries. These memories correspond to progressively coarser views of the week.

At inference time, EgoRAG performs hierarchical retrieval. Given a question and its reference time, the system first selects a target day from day-level summaries, then predicts a narrower time range from hourly summaries, and finally retrieves the top text segments within that range using the question keywords. For questions containing cues such as "last time," the system instead searches the full prior history under time constraints derived from the question and options. Retrieved clips are converted into evidence cards,



(a) Captioning Stage with EgoGPT **(b) Question Answering Stage with EgoRAG**

Figure 4. **EgoButler architecture.** EgoGPT first converts video and audio clips into first-person captions. EgoRAG then aggregates these captions into a hierarchical memory, retrieves evidence using question-driven keyword and time-range selection, and answers the final multiple-choice query from the recovered evidence.

and an answer model selects the final multiple-choice option from the evidence. This design explicitly decomposes week-long QA into *where to look* and *how to answer*.

5. Experiments

5.1. Setup

We evaluate EgoGPT on established egocentric benchmarks and evaluate EgoButler on EgoLifeQA. For EgoLifeQA, all compared captioning models operate under the same pipeline and use 1 FPS video sampling when captions must be generated, following Table 5. We report benchmark accuracy overall and by category, and we further break down EgoRAG performance by certificate length to measure long-horizon behavior.

5.2. Results on Existing Egocentric Benchmarks

Table 4 compares EgoGPT with strong commercial and open-source baselines on EgoSchema, EgoPlan, and EgoThink. EgoGPT trained on EgoIT reaches 73.2 on EgoSchema, and Day-1 personalization further improves it to 75.4, which is the best result in this comparison. The picture is more mixed on the other two benchmarks. EgoPlan improves modestly from 32.4 to 33.4 with personalization but remains below GPT-4V's 38.0. EgoThink slightly drops from 61.7 to 61.4 after personalization and trails the strongest proprietary models. These results suggest that EgoGPT transfers well to long-form egocentric recall, but Day-1 personalization is not uniformly helpful across tasks.

Table 4. **Performance of EgoGPT.** The table compares EgoGPT with strong commercial and open-source models on established egocentric benchmarks. Best results are in bold.

Model	#Param	#Frames	EgoSchema	EgoPlan	EgoThink
GPT-4v [37]	-	32	56.6	**38.0**	**65.5**
Gemini-1.5-Pro [25]	-	32	72.2	31.3	62.4
GPT-4o [24]	-	32	72.2	32.8	**65.5**
LLaVA-Next-Video [20]	7B	32	49.7	29.0	40.6
LongVA [21]	7B	32	44.1	29.9	48.3
IXC-2.5 [38]	7B	32	54.6	29.4	56.0
InternVideo2 [39]	8B	32	55.2	27.5	43.9
Qwen2-VL [23]	7B	32	66.7	34.3	59.3
Oryx [22]	7B	32	56.0	33.2	53.1
LLaVA-OV [19]	7B	32	60.1	30.7	54.2
LLaVA-Videos [40]	7B	32	57.3	33.6	56.4
EgoGPT (EgoIT)	7B	32	73.2	32.4	61.7
EgoGPT (EgoIT+EgoLifeD1)	7B	32	**75.4**	33.4	61.4

5.3. Results on EgoLifeQA

Table 5 reports the main benchmark comparison on EgoLifeQA. Without personalization, EgoGPT reaches 33.1 average accuracy. Adding Day-1 personalization raises this to 36.0, narrowing the gap to GPT-4o (36.2) and Gemini-1.5-Pro (36.9). The gains are not uniform across task types. The largest improvements appear in EntityLog (35.2 to 39.2) and RelationMap (29.6 to 33.6), which is consistent with the idea that user-specific identities and recurring social patterns benefit from personalized memory. EventRecall remains unchanged at 36.5, and HabitInsight improves only modestly from 27.9 to 31.1. GPT-4o remains strongest on EventRecall and TaskMaster, while Gemini-1.5-Pro performs best on

HabitInsight and obtains the best overall average.

6. Analysis

6.1. When Hierarchical Retrieval Helps

Table 6 isolates EgoRAG by grouping questions according to certificate length. The key pattern is clear: hierarchical retrieval mainly helps when the required history is long. Relative to EgoGPT alone, EgoGPT+EgoRAG improves from 29.1 to 35.7 on 2–6 hour questions, from 26.8 to 38.9 on 6–24 hour questions, and from 25.0 to 35.4 on questions requiring more than 24 hours of context. On the shortest bucket, however, performance slightly decreases from 28.2 to 27.2. This is a useful negative result. Retrieval is not a universal improvement; it pays off when the search space is genuinely large, but it can introduce overhead or noise when the answer is already local.

6.2. How Much Caption Quality Matters

Table 7 studies how the caption source affects EgoButler. Human-authored visual-audio captions yield 45.5 average accuracy, substantially above narration alone (31.5) or transcripts alone (29.6). Among model-generated captions, visual-only captions are consistently stronger than audio-only captions, but the best model-generated setup is the full visual+audio EgoGPT with Day-1 personalization at 36.0. The gap between 36.0 and 45.5 shows that retrieval is still bottlenecked by caption quality. Audio helps, especially for speech-heavy categories, but good multimodal grounding is more important than audio alone.

6.3. What EgoLifeQA Reveals

The benchmark statistics in Figure 3 reveal three useful diagnostic properties. First, the task distribution is broad rather than dominated by a single question style: EntityLog, EventRecall, and RelationMap each contribute roughly one quarter of the benchmark, while HabitInsight and TaskMaster probe more specialized assistant capabilities. Second, the benchmark is genuinely long-horizon: two thirds of questions require more than two hours of look-back, and the hardest bins are precisely where retrieval makes the largest difference. Third, the task categories stress different capabilities. RelationMap benefits strongly from personalization, suggesting that identity grounding is a core challenge, whereas HabitInsight remains hard even for stronger models, indicating that extracting durable routines from noisy daily-life observations remains unresolved. Taken together, these patterns suggest that EgoLifeQA is not merely harder because it is longer. If the benchmark difficulty were dominated by generic ambiguity, retrieval and personalization would tend to help uniformly across buckets and task families. Instead, their effects are selective: retrieval helps mainly when the temporal search space becomes large, while personalization helps most when stable identities and social relations matter. This makes EgoLifeQA diagnostically useful for separating failures of temporal search, user modeling, and multimodal perception.

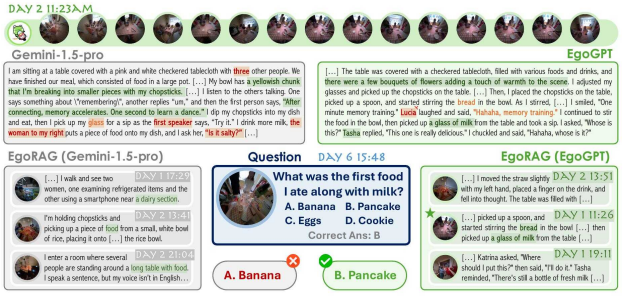


Figure 5. **Qualitative comparison.** Under the EgoButler framework, EgoGPT retrieves more specific evidence than Gemini-1.5-Pro for long-horizon questions, especially when the answer depends on named entities and cross-day recall.

Figure 5 illustrates this behavior qualitatively. EgoGPT-driven retrieval recovers evidence that is more specific about food items, names, and cross-day context, whereas the competing pipeline often retrieves semantically related but less decisive segments. This supports the quantitative result that better egocentric captioning and identity-aware adaptation improve downstream long-context QA.

7. Conclusion

We presented EgoLife, a week-long egocentric dataset for continuous daily-life understanding, EgoLifeQA, a benchmark for personalized long-context assistant queries, and EgoButler, a retrieval-augmented system that combines omni-modal captioning with hierarchical memory retrieval. The experiments show that the benchmark is challenging, that Day-1 personalization mainly helps identity- and relation-centric reasoning, and that hierarchical retrieval is most valuable once questions require many hours of history. At the same time, the results expose important gaps: short-context questions do not automatically benefit from retrieval, and current model-generated captions remain notably weaker than human visual-audio descriptions. We view EgoLifeQA as both an evaluation target and a diagnostic tool for the next generation of egocentric life assistants.

Table 5. **Performance comparison of EgoGPT with strong baselines on the EgoLifeQA benchmark.** For a fair comparison on EgoLifeQA, EgoGPT is replaced with the corresponding model inside the EgoButler pipeline and evaluated under the same conditions. Models that provide captions for EgoLifeQA use 1 FPS video sampling. Best results are in bold.

Model	#Frames	Audio	Identity	EgoLifeQA					
				EntityLog	EventRecall	HabitInsight	RelationMap	TaskMaster	Average
Gemini-1.5-Pro [25]	-	✓	✗	36.0	37.3	**45.9**	30.4	34.9	**36.9**
GPT-4o [24]	1 FPS	✗	✗	34.4	**42.1**	29.5	30.4	**44.4**	36.2
LLaVA-OV [19]	1 FPS	✗	✗	36.8	34.9	31.1	22.4	28.6	30.8
EgoGPT (EgoIT-99K)	1 FPS	✓	✗	35.2	36.5	27.9	29.6	36.5	33.1
EgoGPT (EgoIT-99K+D1)	1 FPS	✓	✓	**39.2**	36.5	31.1	**33.6**	39.7	36.0

Table 6. **Effectiveness of EgoRAG.** Integrating EgoRAG substantially improves long-context question answering, especially for questions with longer certificate lengths. For comparison, we evaluate Gemini-1.5-Pro and EgoGPT on a half-hour video segment, limiting their answers to that local context. Best results are in bold.

Model	Certificate Length			
	$< 2h$	$2h - 6h$	$6h - 24h$	$> 24h$
Gemini-1.5-Pro	27.9	14.8	25.0	18.4
EgoGPT	**28.2**	29.1	26.8	25.0
EgoGPT+EgoRAG	27.2	**35.7**	**38.9**	**35.4**

Table 7. **Caption-source ablation for EgoButler.** We construct different EgoRAG memory banks using captions from EgoGPT variants. The first three rows use human annotations as a reference. All response generation models use EgoGPT (EgoIT-99K+D1) for fair comparison. The results show how caption quality affects downstream EgoButler performance. Best results are in bold.

Caption Source	Visual	Audio	Dataset	Avg.
Narration	✓	✓	-	31.5
Transcript	✗	✓	-	29.6
Visual-Audio Caption	✓	✓	-	**45.5**
EgoGPT (Audio Only)	✗	✓	EgoIT-99K	27.2
EgoGPT (Audio Only)	✗	✓	EgoIT-99K+D1	28.1
EgoGPT (Visual Only)	✓	✗	EgoIT-99K	31.2
EgoGPT (Visual Only)	✓	✗	EgoIT-99K+D1	33.6
EgoGPT (Visual+Audio)	✓	✓	EgoIT-99K	33.1
EgoGPT (Visual+Audio)	✓	✓	EgoIT-99K+D1	36.0

References

[1] Dima Damen, Hazel Doughty, Giovanni Maria Farinella, Sanja Fidler, Antonino Furnari, Evangelos Kazakos, Davide Moltisanti, Jonathan Munro, Toby Perrett, et al. The epic-kitchens dataset: Collection, challenges and baselines. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 43(11):4125–4141, 2021. 1, 2, 3, 4

[2] Kristen Grauman, Andrew Westbury, Eugene Byrne, Zachary Chavis, Antonino Furnari, Rohit Girdhar, Jackson Hamburger, Hao Jiang, Miao Liu, et al. Ego4d: Around the world in 3,000 hours of egocentric video. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 18995–19012, June 2022. 3, 4

[3] Kristen Grauman, Andrew Westbury, Lorenzo Torresani, Kris Kitani, Jitendra Malik, Triantafyllos Afouras, Kumar Ashutosh, Vijay Baiyya, Siddhant Bansal, et al. Ego-exo4d: Understanding skilled human activity from first- and third-person perspectives. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19383–19400, June 2024. 1, 2, 3

[4] Adrián Núñez-Marcos, Gorka Azkune, and Ignacio Arganda-Carreras. Egocentric vision-based action recognition: A survey. *Neurocomputing*, 472:175–197, 2022. 1

[5] Chiara Plizzari, Gabriele Goletto, Antonino Furnari, Siddhant Bansal, Francesco Ragusa, Giovanni Maria Farinella, Dima Damen, and Tatiana Tommasi. An outlook into the future of egocentric vision. *International Journal of Computer Vision*, pages 1–57, 2024. 1

[6] Karttikeya Mangalam, Raiymbek Akshulakov, and Jitendra Malik. Egoschema: A diagnostic benchmark for very long-form video language understanding. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2023. 1, 2, 3, 4

[7] Yi Chen, Yuying Ge, Yixiao Ge, Mingyu Ding, Bohao Li, Rui Wang, Ruifeng Xu, Ying Shan, and Xihui Liu. Egoplan-bench: Benchmarking egocentric embodied planning with multimodal large language models. *arXiv preprint arXiv:2312.06722*, 2023. 3

[8] Sijie Cheng, Zhicheng Guo, Jingwen Wu, Kechen Fang, Peng Li, Huaping Liu, and Yang Liu. Egothink: Evaluating first-person perspective thinking capability of vision-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14291–14302, 2024. 3

[9] Hanrong Ye, Haotian Zhang, Erik Daxberger, Lin Chen, Zongyu Lin, Yanghao Li, Bowen Zhang, Haoxuan You, Dan Xu, Zhe Gan, et al. Mm-ego: Towards building egocentric multimodal llms. *arXiv preprint arXiv:2410.07177*, 2024. 1, 2, 3

[10] Keshigeyan Chandrasegaran, Agrim Gupta, Lea M. Hadzic, Taran Kota, Jimming He, Cristóbal Eyzaguirre, Zane Durante, Manling Li, Jiajun Wu, and Li Fei-Fei. Hourvideo: 1-hour video-language understanding, 2024. 1, 3

[11] Haoning Wu, Dongxu Li, Bei Chen, and Junnan Li. Longvideobench: A benchmark for long-context interleaved video-language understanding. *arXiv preprint arXiv:2407.15754*, 2024. 1, 2

[12] Yifei Huang, Guo Chen, Jilan Xu, Mingfang Zhang, Lijin Yang, Baoqi Pei, Hongjie Zhang, Dong Lu, Yali Wang, et al. Egoexolearn: A dataset for bridging asynchronous ego- and exo-centric view of procedural activities in real world. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. 2, 3

[13] Amir Bar, Arya Bakhtiar, Danny Tran, Antonio Loquercio, Jathushan Rajasegaran, Yann LeCun, Amir Globerson, and Trevor Darrell. Egopet: Egomotion and interaction data from an animal's perspective. *arXiv preprint arXiv:2404.09991*, 2024. 2, 3

[14] Junhao Pan, Zehua Yuan, Xiaofan Zhang, and Deming Chen. Youhome system and dataset: Making your home know you better. *IEEE International Symposium on Smart Electronic Systems (IEEE - iSES)*, 2022. 2, 4

[15] Gunnar A. Sigurdsson, Abhinav Gupta, Cordelia Schmid, Ali Farhadi, and Karteek Alahari. Charades-ego: A large-scale dataset of paired third and first person videos, 2018. 4

[16] Yin Li, Miao Liu, and James M. Rehg. In the eye of beholder: Joint learning of gaze and actions in first person video. In *European Conference on Computer Vision (ECCV)*, 2018. 4

[17] Xin Wang, Taein Kwon, Mahdi Rad, Bowen Pan, Ishani Chakraborty, Sean Andrist, Dan Bohus, Ashley Feniello, Bugra Tekin, Felipe Vieira Frujeri, et al. Holoassist: an egocentric human interaction dataset for interactive ai assistants in the real world. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 20270–20281, 2023. 2, 4

[18] Chaoyou Fu, Yuhan Dai, Yondong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. *arXiv preprint arXiv:2405.21075*, 2024. 2

[19] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. Llava-onevision: Easy visual task transfer. *arXiv preprint arXiv:2408.03326*, 2024. 2, 4, 5, 7

[20] Yuanhan Zhang, Bo Li, haotian Liu, Yong jae Lee, Liangke Gui, Di Fu, Jiashi Feng, Ziwei Liu, and Chunyuan Li. Llava-next: A strong zero-shot video understanding model, April 2024. 5

[21] Peiyuan Zhang, Kaichen Zhang, Bo Li, Guangtao Zeng, Jingkang Yang, Yuanhan Zhang, Ziyue Wang, Haoran Tan, Chunyuan Li, and Ziwei Liu. Long context transfer from language to vision. *arXiv preprint arXiv:2406.16852*, 2024. 5

[22] Zuyan Liu, Yuhao Dong, Ziwei Liu, Winston Hu, Jiwen Lu, and Yongming Rao. Oryx mllm: On-demand spatial-temporal understanding at arbitrary resolution. *arXiv preprint arXiv:2409.12961*, 2024. 5

[23] Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, Yichong Leng, Yuanjun Lv, Jinzheng He, Junyang Lin, et al. Qwen2-audio technical report. *arXiv preprint arXiv:2407.10759*, 2024. 5

[24] OpenAI. Gpt-4o system card, 2024. 3, 5, 7

[25] Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023. 2, 5, 7

[26] Shraman Pramanick, Yale Song, Sayan Nag, Kevin Qinghong Lin, Hardik Shah, Mike Zheng Shou, Rama Chellappa, and Pengchuan Zhang. Egovlpv2: Egocentric video-language pre-training with fusion in the backbone. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 5285–5297, 2023. 2

[27] Lorenzo Baraldi, Costantino Grana, and Rita Cucchiara. Hierarchical boundary-aware neural encoder for video captioning. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1657–1666, 2017. 2

[28] Jie Lei, Liwei Wang, Yelong Shen, Dong Yu, Tamara L Berg, and Mohit Bansal. Mart: Memory-augmented recurrent transformer for coherent video paragraph captioning. *arXiv preprint arXiv:2005.05402*, 2020.

[29] Md Mohaiminul Islam, Ngan Ho, Xitong Yang, Tushar Nagarajan, Lorenzo Torresani, and Gedas Bertasius. Video recap: Recursive captioning of hour-long videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 18198–18208, 2024.

[30] Xiaoqian Shen, Yunyang Xiong, Changsheng Zhao, Lemeng Wu, Jun Chen, Chenchen Zhu, Zechun Liu, Fanyi Xiao, Balakrishnan Varadarajan, Florian Bordes, et al. Longvu: Spatiotemporal adaptive compression for long video-language understanding. *arXiv:2410.17434*, 2024. 2

[31] Jakob Engel, Kiran Somasundaram, Michael Goesele, Albert Sun, Alexander Gamino, Andrew Turner, Arjang Talattof, Arnie Yuan, Bilal Souti, Brighid Meredith, et al. Project aria: A new tool for egocentric multimodal ai research. *arXiv preprint arXiv:2308.13561*, 2023. 3

[32] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *International conference on machine learning*, pages 28492–28518. PMLR, 2023. 3

[33] Mahmoud Ashraf. Whisper diarization: Speaker diarization using openai whisper. 2024. 3

[34] Tim J Schoonbeek, Tim Houben, Hans Onvlee, Fons van der Sommen, et al. Industreal: A dataset for procedure step recognition handling execution errors in egocentric videos in an industrial-like setting. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 4365–4374, 2024. 4

[35] Baoxiong Jia, Ting Lei, Song-Chun Zhu, and Siyuan Huang. Egotaskqa: Understanding human tasks in egocentric videos. *Advances in Neural Information Processing Systems*, 35:3343–3360, 2022. 4

[36] Siddhant Bansal, Chetan Arora, and C.V. Jawahar. My view is the best view: Procedure learning from egocentric videos. In *European Conference on Computer Vision (ECCV)*, 2022. 4

[37] OpenAI. Gpt-4v(ision) system card, 2023. 5

[38] Pan Zhang, Xiaoyi Dong, Bin Wang, Yuhang Cao, Chao Xu, Linke Ouyang, Zhiyuan Zhao, Haodong Duan, Songyang Zhang, Shuangrui Ding, et al. Internlm-xcomposer: A vision-language large model for advanced text-image comprehension and composition. *arXiv preprint arXiv:2309.15112*, 2023. 5

[39] Yi Wang, Kunchang Li, Xinhao Li, Jiashuo Yu, Yinan He, Guo Chen, Baoqi Pei, Rongkun Zheng, Jilan Xu, Zun Wang, et al. Internvideo2: Scaling video foundation models for multimodal video understanding. *arXiv preprint arXiv:2403.15377*, 2024. 5

[40] Yuanhan Zhang, Jinming Wu, Wei Li, Bo Li, Zejun Ma, Ziwei Liu, and Chunyuan Li. Video instruction tuning with synthetic data. *arXiv preprint arXiv:2410.02713*, 2024. 5